Medtronic
Engineering the extraordinary

1.5T and 3T MRI checklist

SureScan™ Pacing, Defibrillation, and CRT-D/CRT-P Systems

Australia & New Zealand

Cardiology Checklist

The following items to be confirmed prior to booking an MRI scan for the patient:

☐ **I confirm the following system requirements are met:**

- Full SureScan™ system implanted (see Cardiac Device Summary Chart over page).
- Pectoral implant and right ventricle location only for Micra™.
- Post lead maturation period complete (approx. 6wks).
- No implanted lead extenders, adaptors, or abandoned leads.

☐ **To my knowledge, at the time of last device check, the following requirements have been met (these parameters will be confirmed at the time of MRI scan, prior to SureScan™ mode being programmed ON):**

- RV pacing threshold
  - Programmed outputs must be ≤ to 4.5V at programmed pulse width for Micra™.
- Lead impedance is within range
  - No broken leads or leads with intermittent electrical contact, as confirmed by lead impedance history.
- Atrial and RV leads are configured to pace Bipolarin all pacemakers and CRT-Ps
- The device is operating within projected service life.
  - Micra model numbers MC1VR01, MC1AVR1, MC2VR01, MC2AVR1 may be scanned if the device is beyond its projected service life AND is programmed to 'Device OFF'

☐ **I confirm that the CareLink SmartSync™ MRI Access app** should be used for this patient. The MRI Access app will automatically determine the appropriate mode and rate. (All Medtronic MR conditional devices are supported EXCEPT Aurora EV-ICD™ or any combination of concomitant IPG/ICD/CRT/Reveal LINQ implanted.)

☐ **No** CareLink SmartSync™ MRI Access app cannot be used with this patient.

MODE (Select one):

☐ AOO ☐ VOO ☐ DOO

☐ ODO/OVO (Rate selection not required for ODO/OVO)

and RATE (Select one. Note: max programmable rate is 120bpm):

☐ 10BPM above average current pre-scan rate (to max rate of ____ bpm)

☐ 20BPM above average current pre-scan rate (to max rate of ____ bpm)

☐ Specific rate (60, 70... 120bpm) _____bpm.

Comments _____

OR:

☐ PLEASE CONTACT ME AT THE TIME OF REPROGRAMMING TO DISCUSS MODE AND RATE.

NOTE:

- For CRT-D devices, RV only pacing is delivered during SureScan™ operation when RV Coil is part of the LV pacing vector AND in all Amplia™, Claria™, Serena™ and Percepta™ devices.
- For a complete set of operating and programming guidelines and restrictions, refer to the respective MRI technical manual for any SureScan™ device.

☐ **I understand patient monitoring will be fulfilled by the radiology team on day of scan as per product labelling.**

☐ **Does this patient require urgent support?**

As the patient's treating cardiologist or as their representative, I am requesting that Medtronic treats this as an emergency case. After evaluating the patient's situation, I have concluded that the patient has a critical medical need for urgent support. I understand this request will result in an increased charge from Medtronic to support this scan.

Physician name:	
Contact number:	
Signature:	
Date:	

[PATIENT LABEL]

1.5T and 3T MRI checklist

SureScan™ Pacing, Defibrillation, and CRT-D/CRT-P Systems

Australia & New Zealand

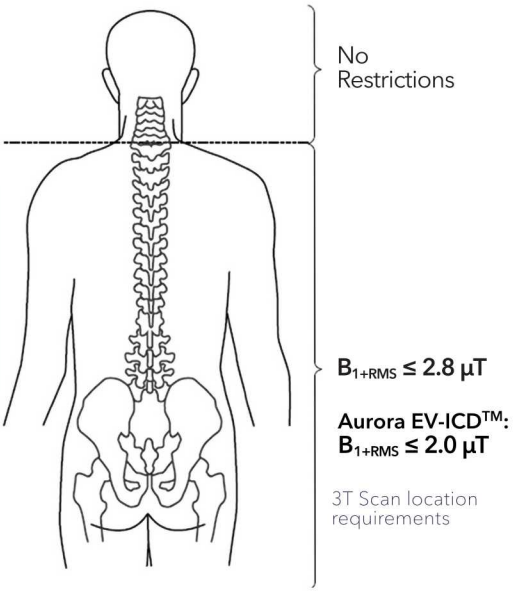
Radiology Checklist

The following items to be confirmed prior to conducting an MRI scan:

☐ A full Medtronic MR-Conditional System confirmed by Cardiologist or Patient Records.

☐ MRI scanner complies with the following:

Scanner type	▪ Horizontal field, cylindrical bore, clinical system for hydrogen proton imaging.
Scanner characteristics	▪ Maximum spatial gradient of ≤ 20 T/m (2,000 gauss/cm). Micra™ ≤ 25T/m (2500 gauss/cm). ▪ Gradient systems with maximum gradient slew rate performance per axis of ≤ 200 T/m/s.



1.5T–MRI radiofrequency (RF) power–Normal Operating Mode

- The whole body averaged specific absorption rate (SAR) must be ≤ 2.0 W/kg.
- Head SAR must be ≤ 3.2 W/kg.
- Micra™ SAR ≤ 4.0 W/kg.

3T–MRI radiofrequency (RF) power

- No restrictions for specific absorption rate (SAR).
- Scans can be performed without restriction when the isocenter (center of the MRI bore) is at or superior to the C7 vertebra (see figure above right).
- B_{1+RMS} must be ≤ 2.8 µT when the isocenter is inferior to the C7 vertebra for all devices except Aurora EV-ICD™ which requires B1+RMS≤ 2.0 µT.
- Micra™ only: there are no patient positioning restrictions relative to the MRI isocenter landmark.

☐ **I confirm the following requirements have been met:**

- Continuous patient monitoring is required while MRI SureScan™ mode is programmed ON.
- All patient monitoring is the responsibility of the radiology team.
- In the event that patient rescue is required, an external defibrillator must be immediately available.

☐ **The patient has consented to Medtronic receiving a copy of this form for the purposes of assisting with the MRI scan if required. Medtronic will hold this information in accordance with the Medtronic Privacy Policy found at www.medtronic.com.au**

☐ **I confirm that MRI SureScan™ mode will be programmed ON prior to the scan and OFF after the scan**

Radiology Staff Name:	
Signature:	
Date:	
Designation:	

[PATIENT LABEL]

MR-Conditional Cardiac Device

Summary Chart

Australia & New Zealand

Verify only Medtronic MR-Conditional Systems, including the following components, are implanted.

Therapy	Product	Model Number	MR Conditional*	MR system	SureScan™ Leads 1.5T or 3T
Pacemakers	Micra™	MC1VR01, MC2VR01, MC1AVR1, MC2AVR1	Yes	Horizontal cylindrical bore magnet 1.5T or 3T	Not Applicable
	Advisa MRI™	A3DR01, A3SR01	Yes – If complete system consists of an implanted SureScan™ pacemaker and SureScan™ lead(s)		Pacing leads: 3830** Lengths: 59, 69, 74 cm 4076, 5076 Lengths: 35, 45, 52, 58, 65, 85 cm 4074 Lengths: 52, 58 cm 4574 Lengths: 45, 53 cm 5086MRI Lengths: 45, 52, 58 cm
	Ensura MRI™	EN1DR01, EN1SR01			
	Azure MRI™	W3DR01, W3SR01, W2DR01, W2SR01			
Cardiac Defibrillators	Visia AF MRI™	DVFB2D1†, DVFB2D4 DVFC3D1†, DVFC3D4	Yes – If complete system is implanted with a SureScan™ ICD and SureScan™ lead(s)	Horizontal cylindrical bore magnet 1.5T or 3T	Defibrillation leads: 6947M, 6935M Lengths: 55, 62 cm 6947, 6935 Lengths: 58, 65 cm EV2401 Lengths: 52, 63 cm See above for pacing leads.
	Evera MRI™	DDMB2D4, DVMB2D4, DDMB2D1†, DVMB2D1† DDMC3D4, DVMC3D4, DDMC3D1†, DVMC3D1†			
	Cobalt™ MRI	DVPA2D1†, DVPA2D4, DDPA2D1†, DDPA2D4 DVPB3D1†, DVPB3D4, DDPB3D1†, DDPB3D4			
	Crome™ MRI	DVPC3D1†, DVPC3D4, DDPC3D1†, DDPC3D4			
	Aurora EV-ICD™	DVEA3E4			
Cardiac ResynchroniSation Therapy Defibrillators	Claria MRI™ CRT-D	DTMA2QQ, DTMA2Q1† DTMA2D4, DTMA2D1†	Yes – If complete system is implanted with a SureScan™ CRT-D and SureScan™ lead(s) or the Model 6725 atrial pin plug	Horizontal cylindrical bore magnet 1.5T or 3T	CRT leads: 4196, 4296, 4396, 4298, 4398, 4598 Lengths: 78, 88 cm 4796, 4798 Lengths: 88 cm See above for defibrillation and pacing leads.
	Amplia MRI™ CRT-D	DTMB2QQ DTMB2D4, DTMB2D1†			
	Cobalt HF™ CRT-D MRI	DTPA2QQ, DTPA2Q1†, DTPA2D4, DTPA2D1† DTPB2QQ, DTPB2Q1†, DTPB2D4, DTPB2D1†			
	Crome HF™ CRT-D MRI	DTPC2QQ, DTPC2Q1†, DTPC2D4, DTPC2D1†			

* Multiple MR-Conditional devices acceptable if MR labelling conditions for all implants can be satisfied. Not all the products listed above are available for supply in Australia and New Zealand. Some devices are listed to cater for patient management considerations where the patients may have these devices still in service.

** SelectSecure 3830 implanted in the conventional RA or RV locations, His Bundle location, left bundle branch area or a combination of these locations is MRI Conditional. A lead other than one of the CRT Leads (as per the above table) plugged into the LV port of any CRT device is NOT MRI Conditional.

† When a single coil SureScan™ defibrillation lead is used, a Medtronic DF-1 pin plug must be secured in the SVC port to make a complete SureScan™ DF-1 defibrillation system.

NB. For EnRhythm MRI please see product manual.

MR-Conditional Cardiac Device

Summary Chart (Cont)

Australia & New Zealand

Therapy	Product	Model Number	MR Conditional*	MR system	SureScan™ Leads 1.5T or 3T
Cardiac ResynchroniSation Therapy Pacemakers	Percepta™ CRT-P	W4TR04, W1TR04	Yes – If complete system is implanted with a SureScan™ CRT-P and SureScan™ lead(s) or the Model 6725 atrial pin plug	Horizontal cylindrical bore magnet 1.5T or 3T	CRT leads: 4196, 4296, 4396, 4298, 4398, 4598 Lengths: 78, 88 cm 4796, 4798 Lengths: 88 cm See above for pacing leads.**
	Serena™ CRT-P	W4TR05, W1TR05			
Insertable Cardiac Monitors	Reveal LINQ™‡	LNQ11, LNQ22	Yes	Horizontal cylindrical bore magnet 1.5T or 3T	Not Applicable
	Reveal™ XT‡	9529	Yes	Closed bore, cylindrical magnet 1.5T or 3T	

*Multiple MR-Conditional devices acceptable if MR labelling conditions for all implants can be satisfied. Not all the products listed above are available for supply in Australia and New Zealand. Some devices are listed to cater for patient management considerations where the patients may have these devices still in service.

** SelectSecure 3830 implanted in the conventional RA or RV locations, His Bundle location, left bundle branch area or a combination of these locations is MRI Conditional. A lead other than one of the CRT Leads (as per the above table) plugged into the LV port of any CRT device is NOT MRI Conditional.

‡ If deemed necessary by Cardiology, review device data pre and/or post scan via usual follow up practice.

NB. For EnRhythm MRI please see product manual.

See the appropriate product MRI SureScan™ Technical Manual before performing an MRI Scan and see the device manuals for detailed information regarding the implant procedure, indications, contraindications, warnings, precautions, and potential complications/adverse events.

To obtain the MR Conditions for use, please visit: www.manuals.medtronic.com

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